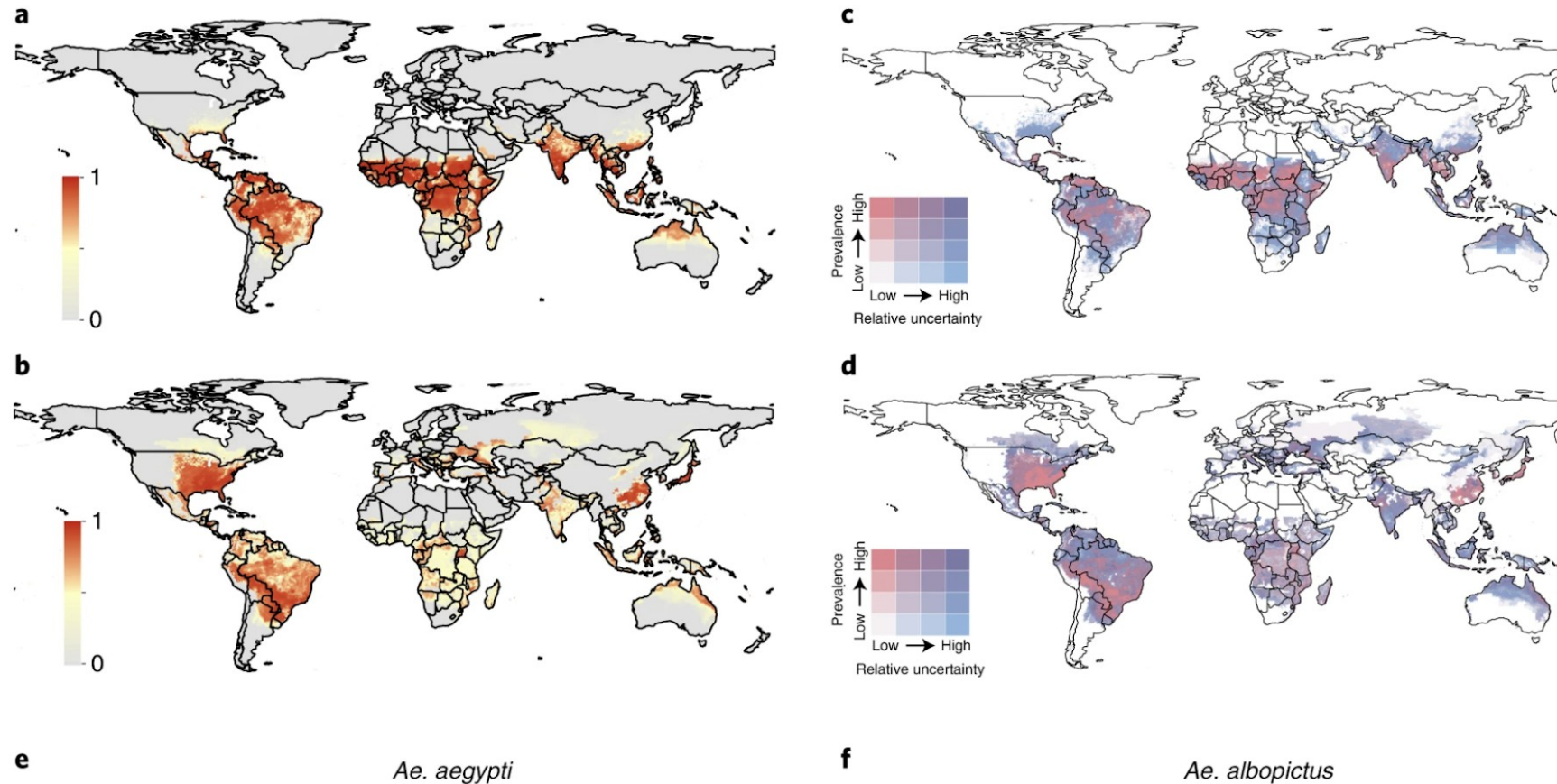


Climate-Sensitive Vector Dynamics  
Modelling Workshop  
September 17-19, 2025 - Nicosia, Cyprus



Summary of open access observational vector presence and abundance datasets

[william.wint@gmail.com](mailto:william.wint@gmail.com)

# DATA NEEDS



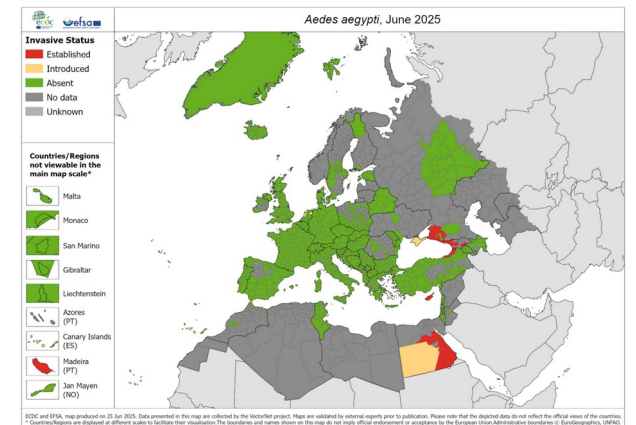
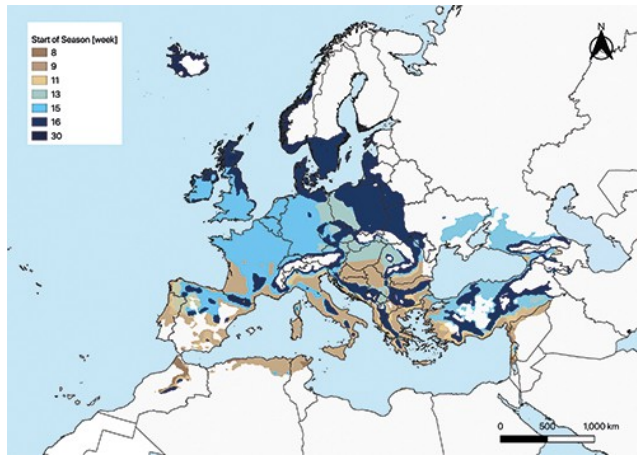
What are *Aedes* data needed for:

Mechanistic population models (validation)  
based on abundance, life stage, traits etc

Spatial distribution models of  
presence/absence  
abundance  
NB also life stage/sex

Seasonal activity  
season start  
season end  
season length

Dissemination and illustration, advocacy  
simplified maps, overviews



Variables:

# DATA VARIABLES



Presence (and absence)

Useful for

seasonality and activity

dependent on complete sampling

Less useful for mechanistic/deterministic models

Absence

Just as important, at least for distribution modelling

Rarely provided

Often need to generate

Seasonality

Start end dates, peak numbers/sample effort

Abundance

?? Related to risk??

How to measure?

? Complete season needed

Standardised indices/Max number

Standardised collection methods

How to equilibrate counts of different life stages

# SOURCES: FIELDWORK



Usually snapshots and limited spatial and temporal coverage,  
may be ok for deterministic modelling as often site specific  
needs to be aggregated for use at scale  
standard metrics essential

VectorNet: **will become more relevant in coming years**

field work over six years, targeted to fill distribution gaps  
standardised sampling strategy  
20+ mosquito species  
historically pres abs , recently abundance with effort



AIMSURV – voluntary - EU **albopictus focused**, designed for abundance  
modelling

42 teams, 24 countries  
standardised ovitrap sampling methodology, two tier sampling strategy  
**Template for aegypti in future**

Many many smaller scale reported in literature (vectorByte) - **global**

# SOURCES: LITERATURE



## Literature

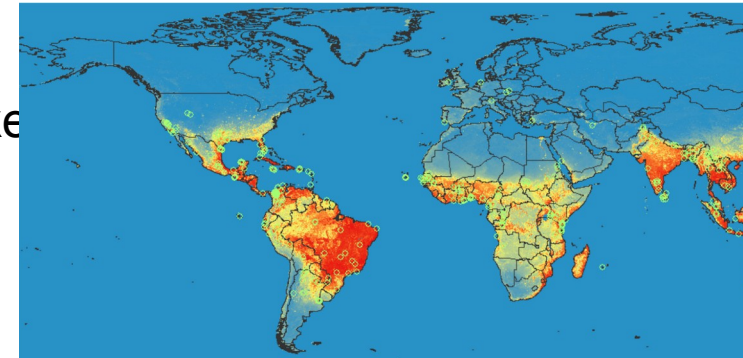
available but very labour intensive  
standardisation and extraction an issue – also publications may be incomplete

**Kraemer *et al*** remains major example but is now old, presence only

e4warning informal update

publish or perish – quicker

programmatic scanning,  
aegypti  
presence only  
easy to make errors



abundance data much more difficult and no comprehensive attempt yet made

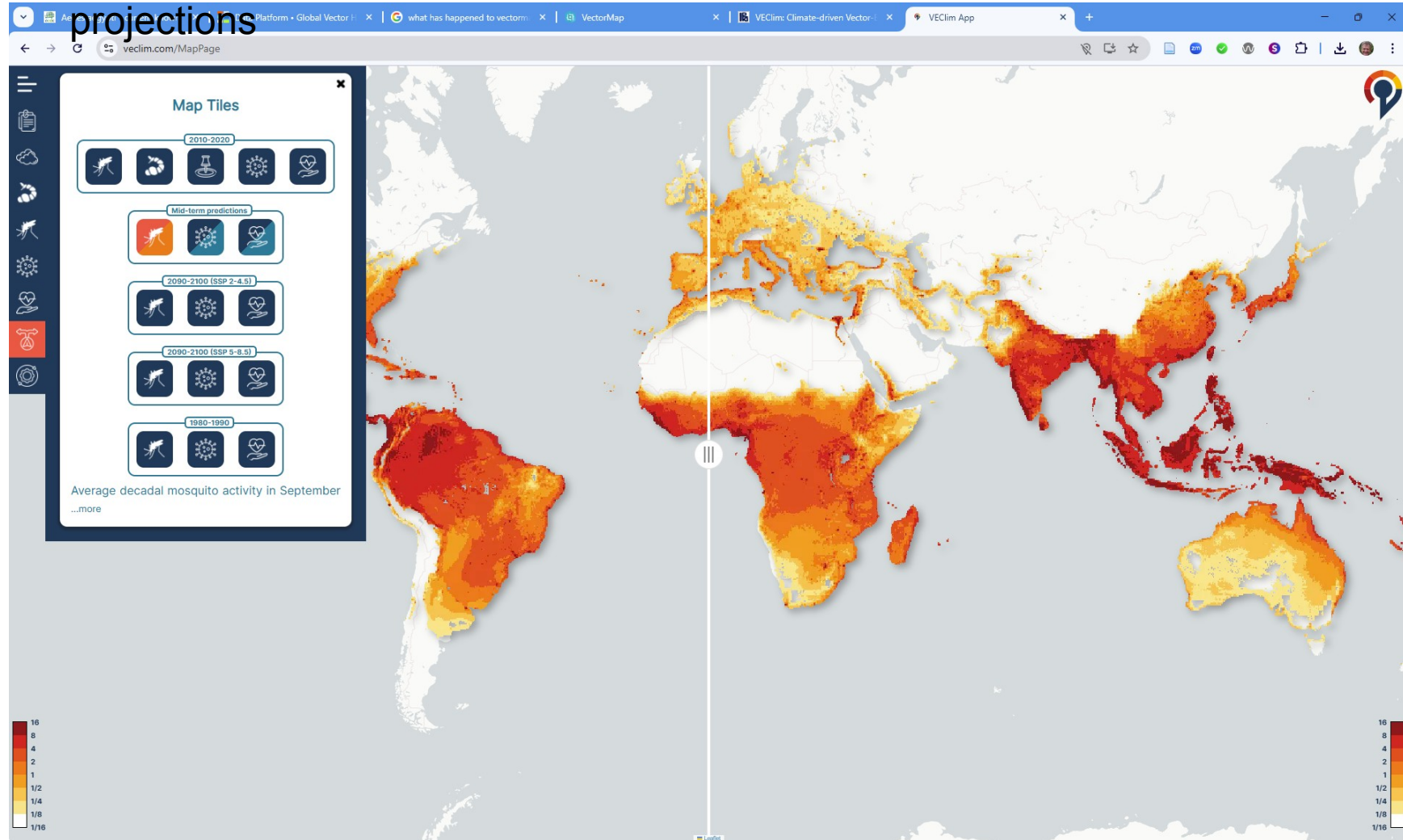
# SOURCES:

easy to get at more difficult to process especially for abundance

# ONLINE



VEclim : albopictus/ detailed lifestage and risk forecasts and  
projections



William Wint, Available *Aedes* datasets, Modelling workshop, Cyprus, Sept 25

# SOURCES:

Vectorbyte., traits (e.g. vectraits 40 aegypti, 50 albo) and abundance database lists



**Characterizing the Vector Data Ecosystem**

A growing body of information on vector-borne diseases has arisen as increasing research focus has been directed towards the need for anticipating risk, optimizing surveillance, and understanding the fundamental biology of vector-borne diseases to direct efforts to control and mitigation. The scope and scale of this information, in the form of data, comprising database efforts, data storage, and serving approaches, mean that it is distributed across many formats and data types. Data ranges from collections records to molecular characterization, geospatial data to interactions of vectors and traits, infection experiments to field trials. New initiatives arise, often spanning the effort traditionally siloed in specific research disciplines, and other efforts wane, perhaps in response to funding declines, different research directions, or lack of sustained interest. Thusly, the world of vector data – the Vector Data Ecosystem – can become unclear in scope, and the flows of data through these various efforts can become stymied by obsolescence, or simply by gaps in access and interoperability. As increasing attention is paid to creating FAIR (Findable Accessible Interoperable, and Reusable) data, simply characterizing what is ‘out there’, and how these existing data aggregation and collection efforts interact, or interoperate with each other, is a useful exercise. This website and related project presents a snapshot of current vector data efforts, a brief description of their stated scope and purpose, and level of accessibility. We [welcome additions](#) to this set, following a similar template, to improve this resource for the larger vector ecology and vector-borne disease research and practitioner community.

We invite you to [view our blog post](#) about this work and see the [publication](#) on the database which includes more context and commentary.

(FO) Fully Open Database  
(PA) Partially Accessible Database

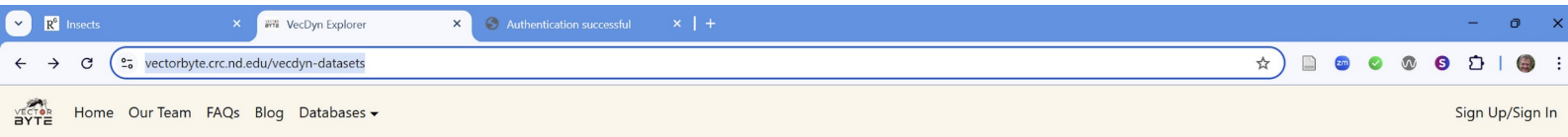
**Global Biological Information Facility (GBIF) depositors and aggregators**

GBIF GBIF.us (Formerly BISON) SCAN

By using this website, you agree to our use of cookies. We use cookies to provide you with a great experience and to help our website run effectively. [Accept](#)

# SOURCES:

VectorByte VEDyn App individual collections of abundance datasets,  
<https://vectorbyte.crc.nd.edu/vecdyn-datasets>



## VecDyn Explorer

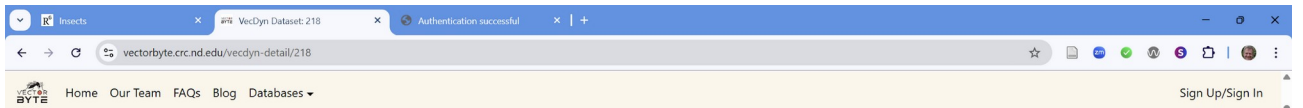
Instructions +

Search by phrase...

Select a column ▾ Select an operator ▾ Enter a search term...

Click any row in the table to view its details page or click the checkbox beside any number of rows and then click the download button to download their data.

| Select                   | Species Name                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Title                                                                                         | Co  |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-----|
| <input type="checkbox"/> | <i>Aedes albopictus</i> , <i>Aedes canadensis</i> complex, <i>Aedes cinereus</i> , <i>Aedes grossbecki</i> , <i>Aedes japonicus</i> complex, <i>Aedes nigromaculis</i> , <i>Aedes sollicitans</i> , <i>Aedes sp.</i> , <i>Aedes sticticus</i> , <i>Aedes thibaulti</i> , <i>Aedes triseriatus</i> , <i>Aedes trivittatus</i> , <i>Aedes vexans</i> , <i>Anopheles barberi</i> , <i>Anopheles crucians</i> , <i>Anopheles punctipennis</i> , <i>Anopheles quadrimaculatus</i> , <i>Anopheles sp.</i> , <i>Anopheles walkeri</i> , BLANK, <i>Coquillettidia perturbans</i> , <i>Culex pipiens</i> morphological group, <i>Culex salinarius</i> , <i>Culex sp.</i> , <i>Culex tarsalis</i> , <i>Culex territans</i> , <i>Culicidae</i> , <i>Culiseta inornata</i> , <i>Culiseta melanura</i> , <i>Culiseta sp.</i> , <i>Psorophora ciliata</i> , <i>Psorophora cyaneescens</i> , <i>Psorophora ferox</i> , <i>Psorophora howardii</i> , <i>Psorophora sp.</i> , <i>Uranotaenia sapphirina</i> | 2023 Mosquito Surveillance St Joseph County Indiana USA                                       | 208 |
| <input type="checkbox"/> | <i>Aedes cinereus</i> , <i>Aedes dorsalis</i> , <i>Aedes fitchii</i> , <i>Aedes increpitus</i> , <i>Aedes melanion</i> , <i>Aedes nigromaculis</i> , <i>Aedes sierrensis</i> , <i>Aedes sticticus</i> , <i>Aedes vexans</i> , <i>Anopheles freeborni</i> , BLANK, <i>Coquillettidia perturbans</i> , <i>Culex pipiens</i> , <i>Culex tarsalis</i> , <i>Culiseta incidens</i> , <i>Culiseta inornata</i> , <i>Culiseta minnesotae</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2021-2023 Ada, Idaho                                                                          | 524 |
| <input type="checkbox"/> | <i>Aedes aegypti</i> , <i>Aedes albopictus</i> , <i>Aedes atlanticus</i> , <i>Aedes infirmatus</i> , <i>Aedes sp.</i> , <i>Aedes taeniorhynchus</i> , <i>Aedes vexans</i> , <i>Anopheles bradleyi</i> , <i>Anopheles crucians</i> complex, <i>Anopheles punctipennis</i> , <i>Anopheles quadrimaculatus</i> , <i>Anopheles sp.</i> , BLANK, <i>Coquillettidia perturbans</i> , <i>Culex erraticus</i> , <i>Culex nigripalpus</i> , <i>Culex quinquefasciatus</i> , <i>Culex sp.</i> , <i>Culiseta melanura</i> , <i>Mansonia genus sp.</i> , <i>Mansonia titillans</i> , <i>Melanoconion Sp.</i> , <i>Psorophora ciliata</i> , <i>Psorophora columbiae</i> , <i>Psorophora ferox</i> , <i>Psorophora sp.</i> , <i>Toxorhynchites rutilus</i> , Unknown unknown, <i>Uranotaenia lowii</i> , <i>Uranotaenia sapphirina</i> , <i>Uranotaenia sp.</i> , <i>Wyeomyia sp.</i>                                                                                                                    | 2010-2019 Collier County                                                                      | 406 |
| <input type="checkbox"/> | <i>Aedes albopictus</i> , <i>Aedes canadensis</i> complex, <i>Aedes cinereus</i> , <i>Aedes hendersoni</i> , <i>Aedes japonicus</i> complex, <i>Aedes sollicitans</i> , <i>Aedes sticticus</i> , <i>Aedes stimulans</i> , <i>Aedes triseriatus</i> , <i>Aedes trivittatus</i> , <i>Aedes vexans</i> , <i>Anopheles barberi</i> , <i>Anopheles punctipennis</i> , <i>Anopheles</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2021-2023 Mosquito Surveillance, Desplaines Valley Mosquito Abatement District (Chicago-land) | 412 |



### Experimental Information

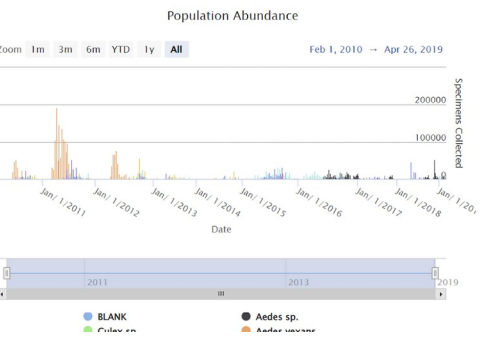
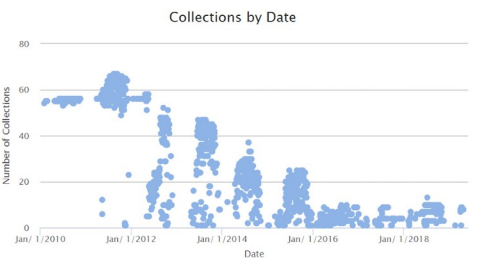
|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dataset Number:             | 218                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Species                     | <i>Aedes aegypti</i> , <i>Aedes albopictus</i> , <i>Aedes atlanticus</i> , <i>Aedes infirmatus</i> , <i>Aedes sp.</i> , <i>Aedes taeniorhynchus</i> , <i>Aedes vexans</i> , <i>Anopheles bradleyi</i> , <i>Anopheles crucians</i> complex, <i>Anopheles punctipennis</i> , <i>Anopheles quadrimaculatus</i> , <i>Anopheles sp.</i> , BLANK, <i>Coquillettidia perturbans</i> , <i>Culex erraticus</i> , <i>Culex nigripalpus</i> , <i>Culex quinquefasciatus</i> , <i>Culex sp.</i> , <i>Culiseta melanura</i> , <i>Mansonia genus sp.</i> , <i>Mansonia titillans</i> , <i>Melanoconion Sp.</i> , <i>Psorophora ciliata</i> , <i>Psorophora columbiae</i> , <i>Psorophora ferox</i> , <i>Psorophora sp.</i> , <i>Toxorhynchites rutilus</i> , Unknown unknown, <i>Uranotaenia lowii</i> , <i>Uranotaenia sapphirina</i> , <i>Uranotaenia sp.</i> , <i>Wyeomyia sp.</i> |
| Years:                      | 2016, 2017, 2018, 2019, 2010, 2011, 2012, 2013, 2014, 2015                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Collection Method(s):       | BG-SENT bg-lure;co2, CDCLIGHT co2;light, NJLT octenol                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Publication Title(s):       | 2010-2019 Collier County                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Publication Description(s): |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Location(s):                | Collier County, Florida, USA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

### Data Source

|              |          |
|--------------|----------|
| Imported by: | Sam Rund |
| Dataset DOI: | No DOI   |

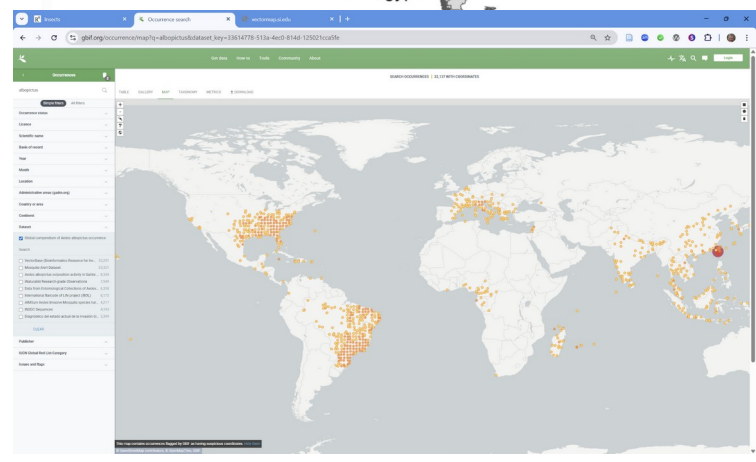
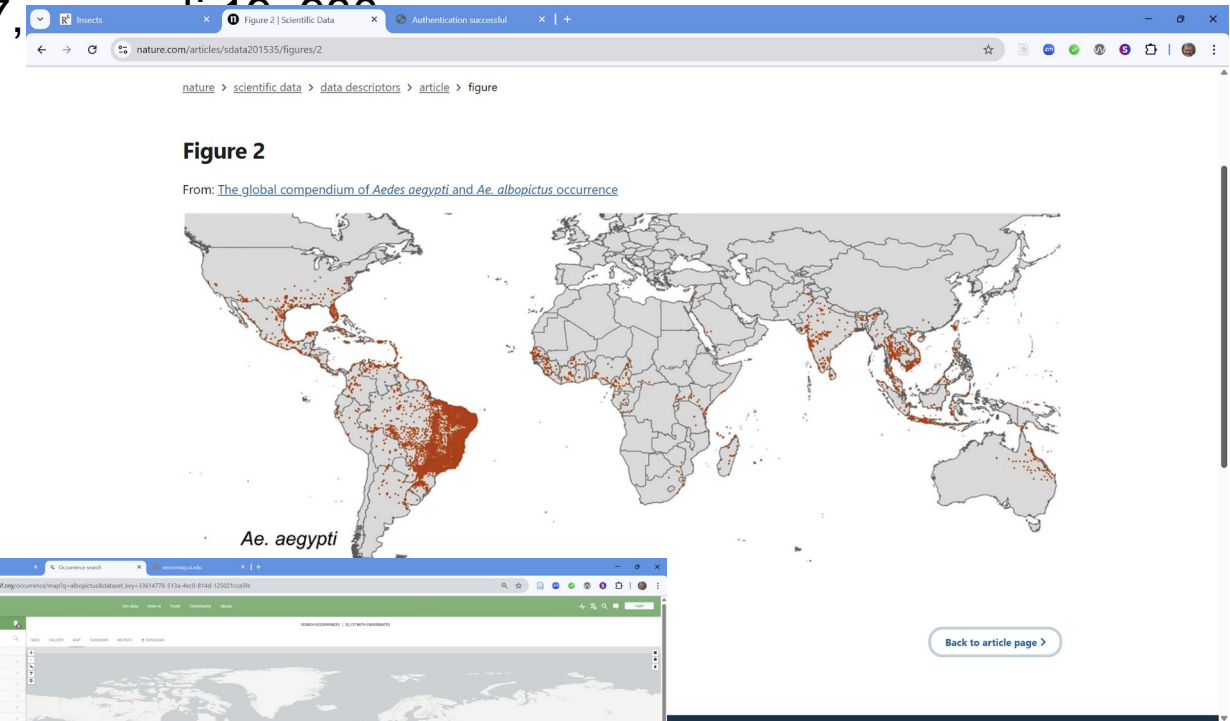
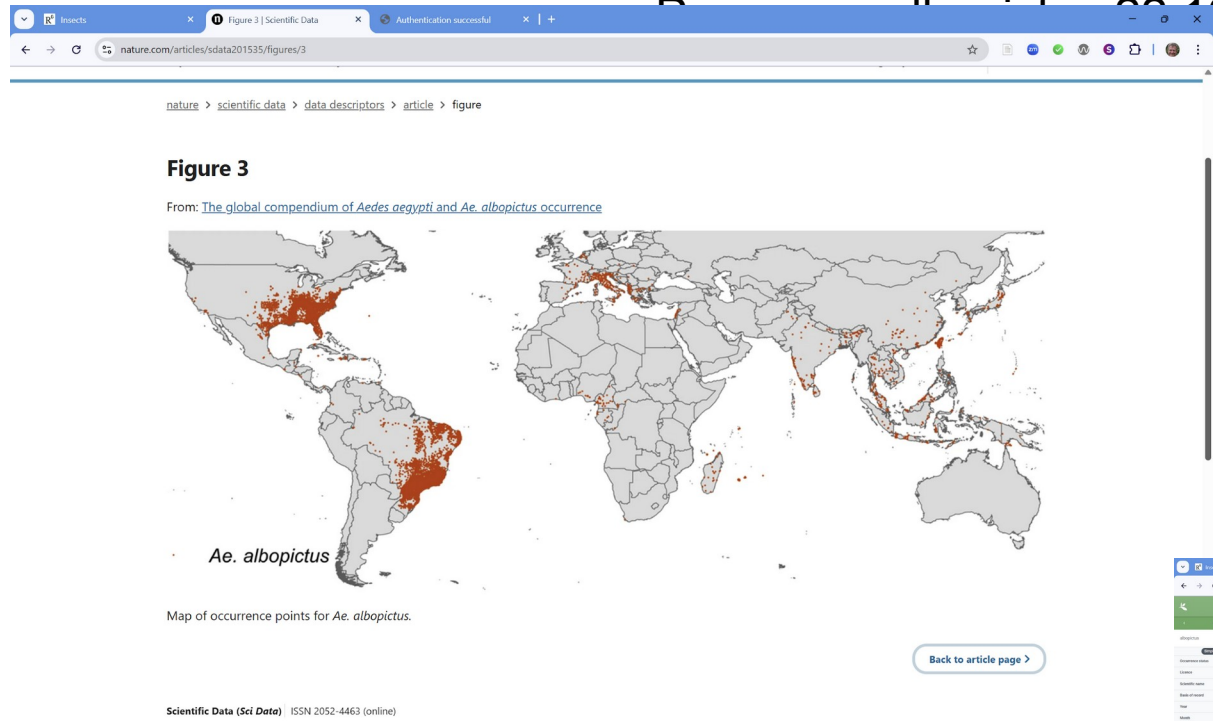
### Data

| Species            | Sample Start Date | Sample End Date | Sample Value | Sample Lat Dd | Sample Long Dd | Sampling Method |
|--------------------|-------------------|-----------------|--------------|---------------|----------------|-----------------|
| Uranotaenia sp.    | 2015-10-05        | 2015-10-05      | 16.0         | 26.4440703    | -81.4314981    | NJLT octenol    |
| Melanoconion Sp.   | 2015-10-05        | 2015-10-05      | 64.0         | 26.4440703    | -81.4314981    | NJLT octenol    |
| Mansonia genus sp. | 2015-10-05        | 2015-10-05      | 8.0          | 26.4440703    | -81.4314981    | NJLT octenol    |
| Culex nigripalpus  | 2015-10-05        | 2015-10-05      | 984.0        | 26.4440703    | -81.4314981    | NJLT octenol    |
| Anopheles sp.      | 2015-10-05        | 2015-10-05      | 304.0        | 26.4440703    | -81.4314981    | NJLT octenol    |





# SOURCES: ONLINE Global compendium, Kraemer et al



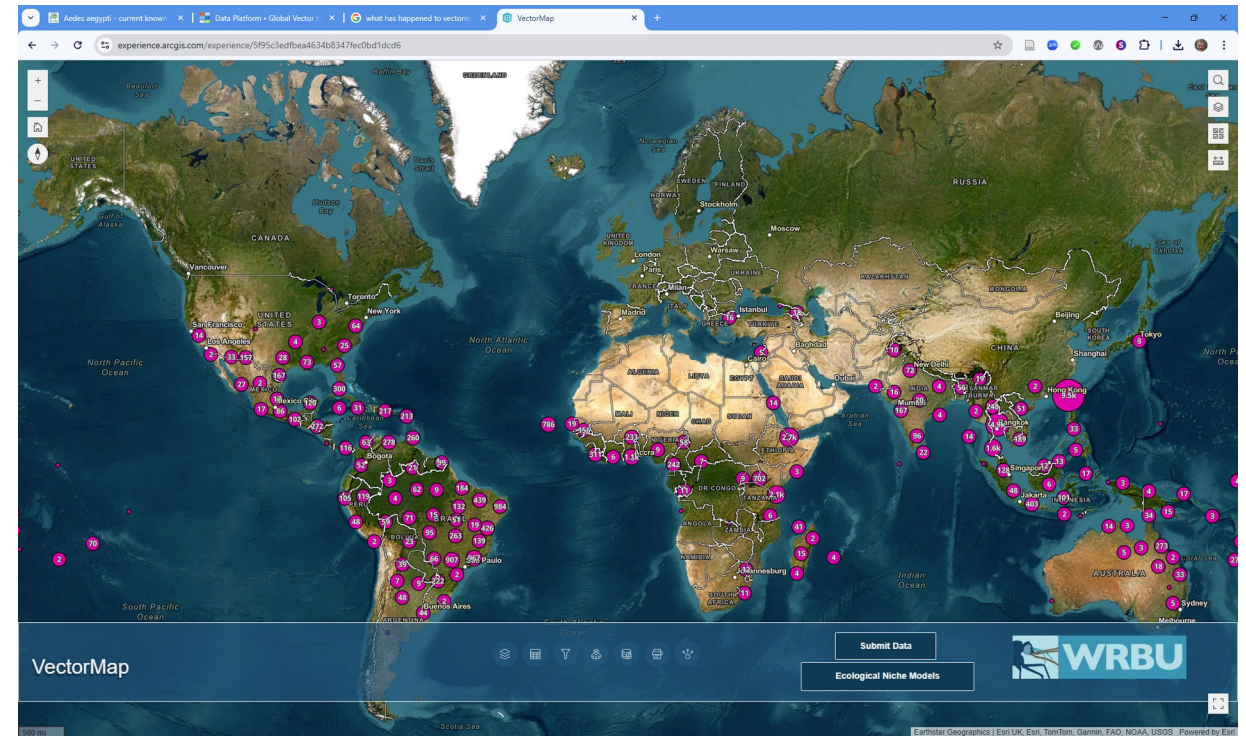
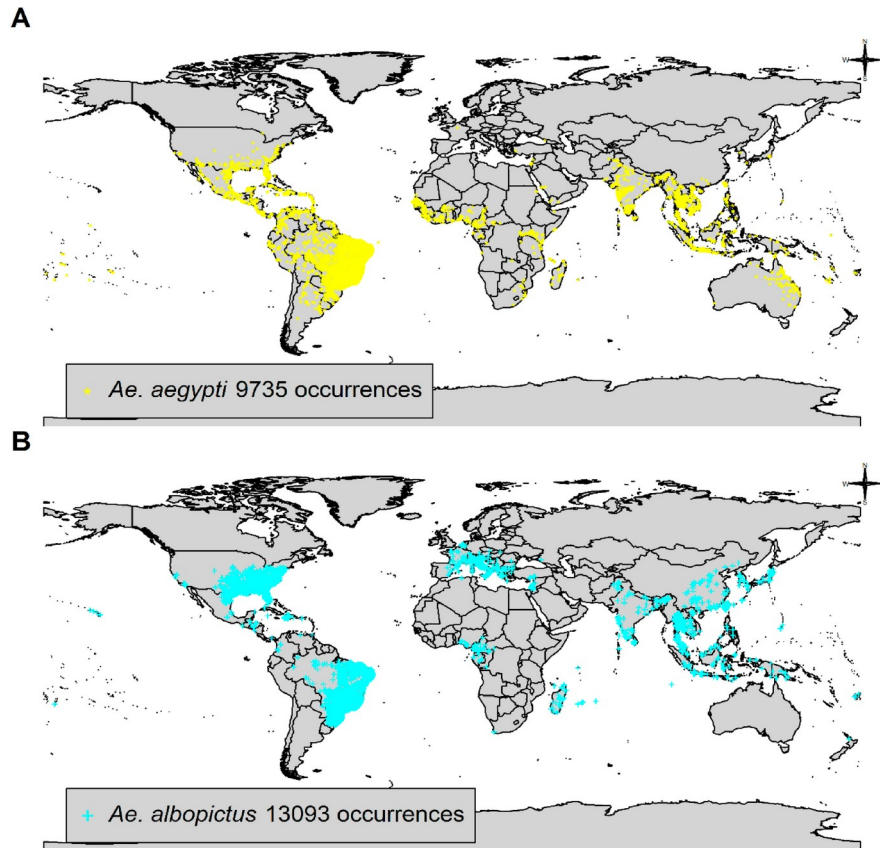
The godfather of them all,  
reformatted by Page et al, for

# SOURCES:



VectorMap: Laporta, G.Z.; Potter, A.M.; Olliaro, J.L.; Bourke, B.P.; Pecor, D.B.; Linton, Y.-M. Global Distribution of *Aedes aegypti* and *Aedes albopictus* in a Climate Change Scenario of Regional Rivalry. *Insects* **2023**, *14*, 49. <https://doi.org/10.3390/insects14010049> (vectormap.si.edu ) now at <https://experience.arcgis.com/experience/5f95c3edf6ea4634b8347fec0bd1dcd6>

*aegypti* 38.3K, not sure about albo pictus, also hosts, bloodmeals, Insect Resistance

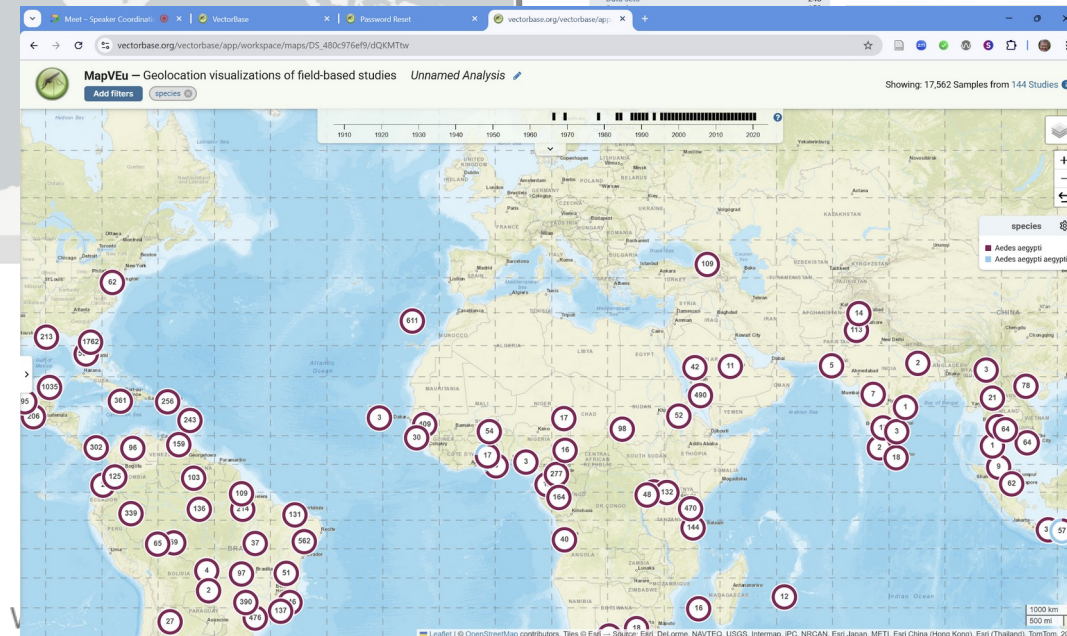
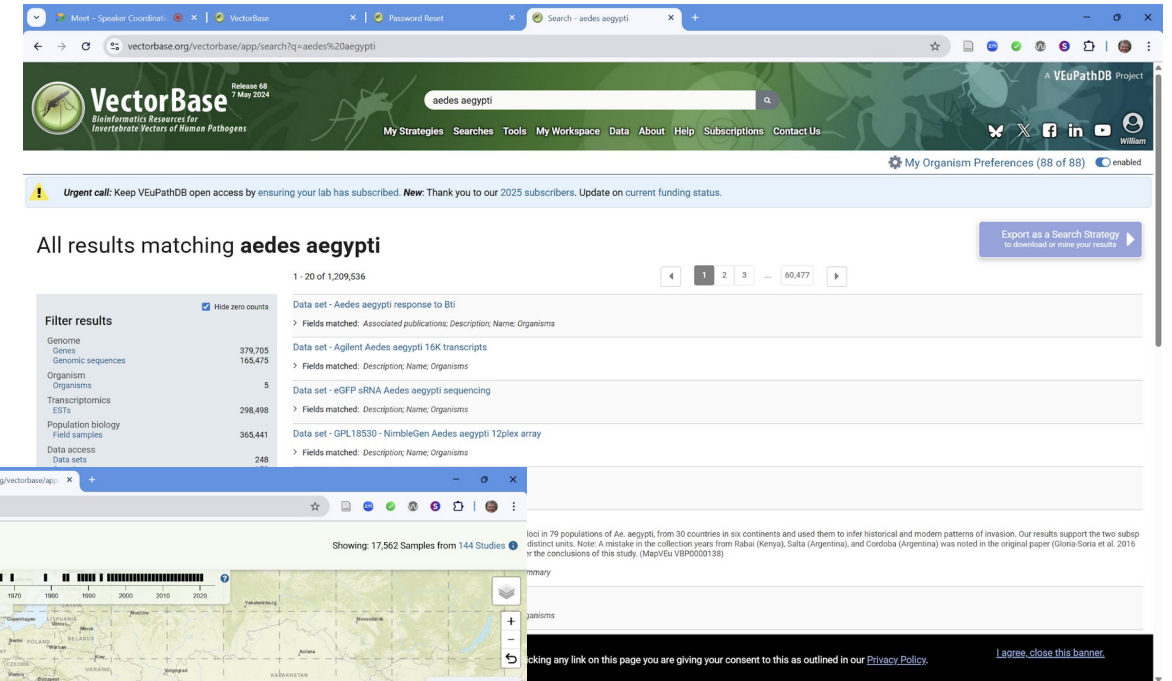
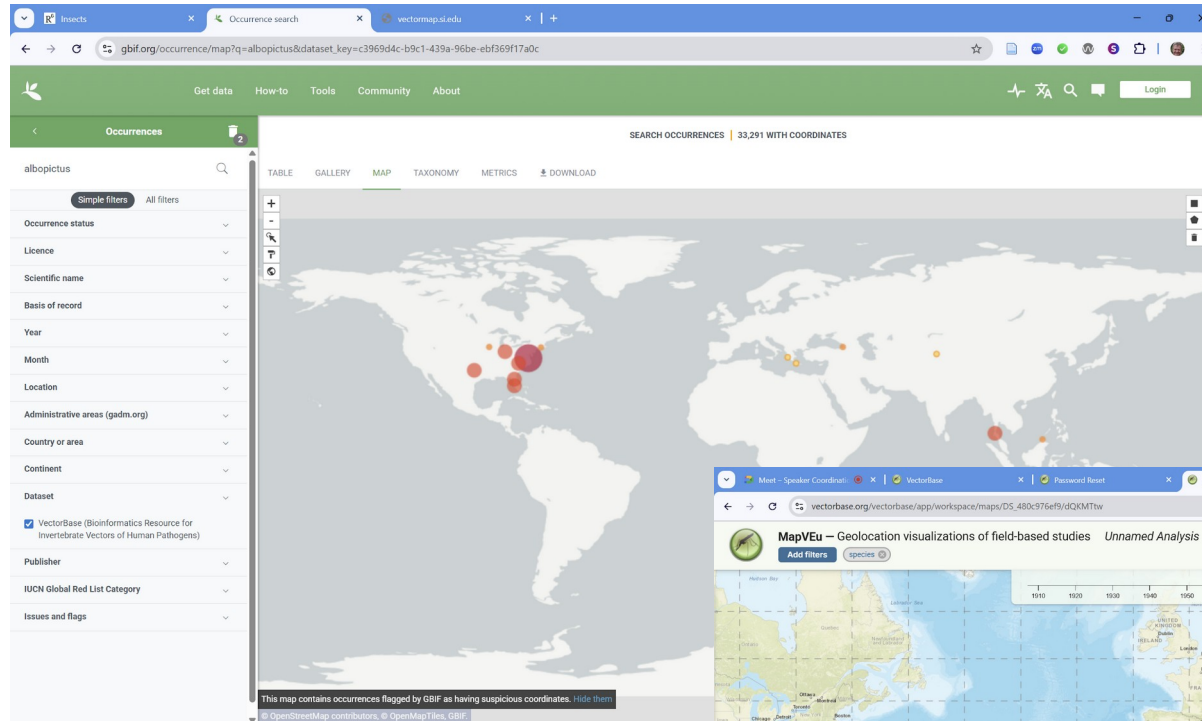


Failed to find any way to  
download

William Wint, Available *Aedes* datasets, Modelling workshop, Cyprus, Sept 25

# SOURCES:

**ONLINE**  
VectorBase <https://vectorbase.org/vectorbase/app>.  
1.6M records, 33,291 albopictus, 14,561 aegypti  
Genome details , **NOW SUBSCRIPTION**

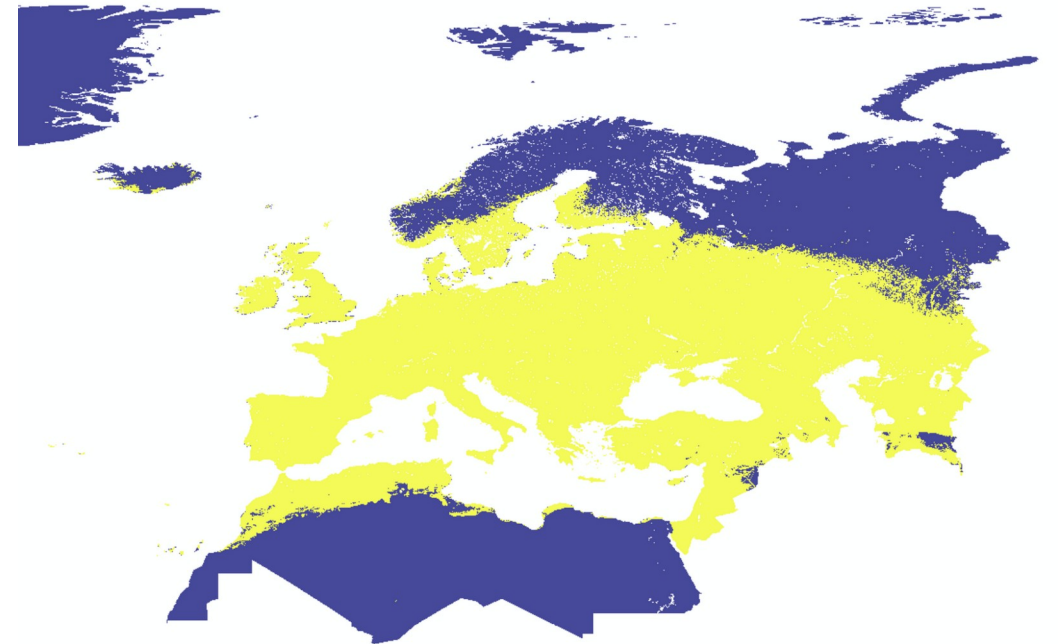
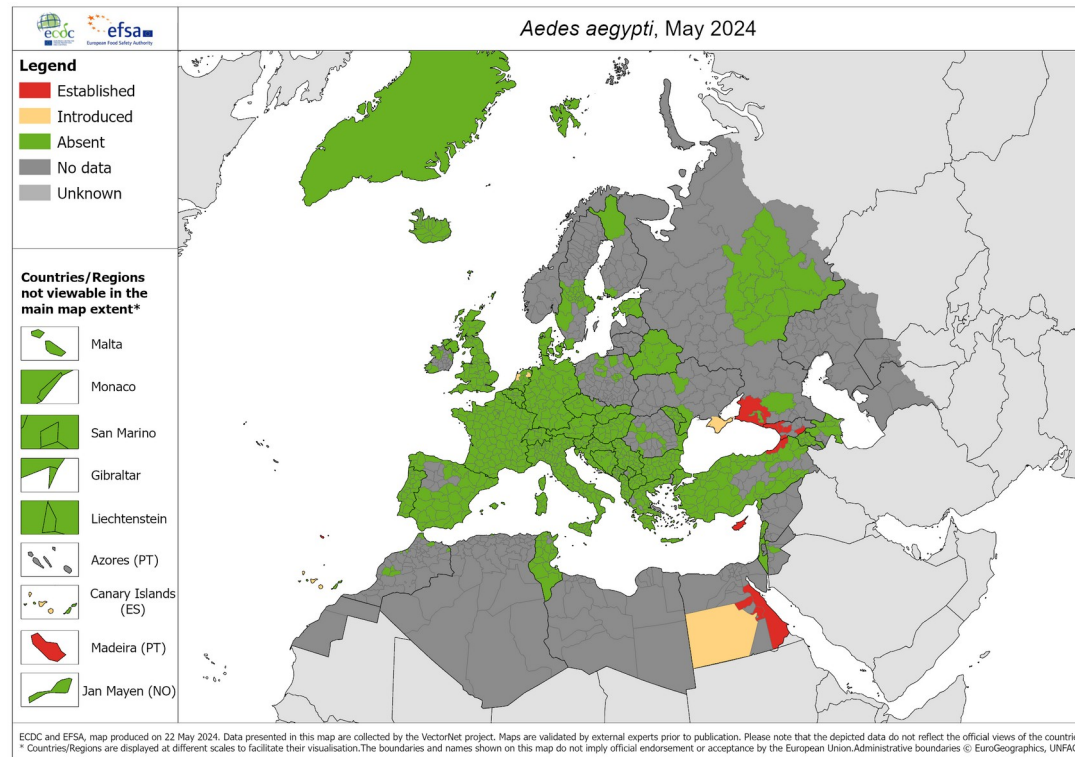


# SOURCES: ONLINE



## VectorNet

Mostly known for status maps  
records with mixture of polygon, point, presence absence and abundance data. Literature and fieldwork. Also (soon) calculated absence from environmental thresholds and also from sampling methodology. To be added to GBIF. 40,239 also, limited number aegypti



## Mosquito Alert

# SOURCES: ONLINE



23K *albopictus* Europe centric no *aegypti* But rapid reaction. and more complete for



MOSQUITO  
ALERT



# SOURCES: ONLINE



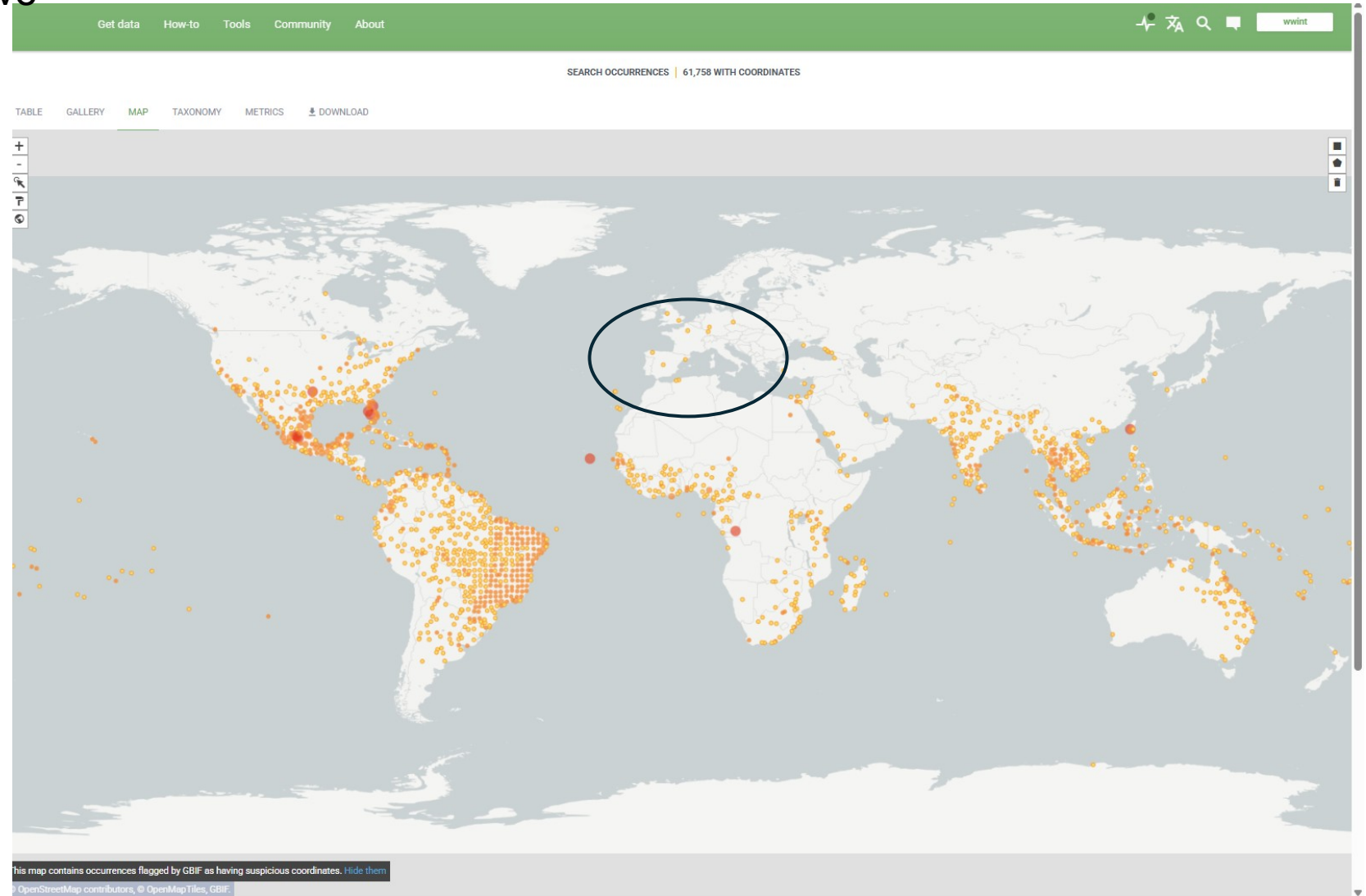
Has sampling effort, burners, derived presence, many duplicates, need to be selective

albopictus 135K,  
aegypti 63K  
Most major datasets. Not  
yet VectorNet

Caveat  
emptor

Search

- ☐ Global compendium of *Aedes aegypti* occ... 19,929
- ☐ VectorBase (Bioinformatics Resource for I... 14,567
- ☐ Apoyo para la infraestructura de la colecció... 6,556
- ☐ Actualización de la Colección de Artrópodo... 6,215
- ☐ iNaturalist Research-grade Observations 3,617
- ☐ Fiocruz/CMN - Coleção de Mosquitos Neot... 3,095
- ☐ INSDC Sequences 3,012
- ☐ *Aedes* mosquito distribution across urban ... 1,939
- ☐ Collection of entomological, demographic, ... 1,920
- ☐ Naturalis Biodiversity Center (NL) - Diptera 1,502



# DATA WRANGLING



## Abundance equilibrations:

|                           | Ovitrap | larval | BG traps |        | CDC traps |        | EVS traps |        | Human   |
|---------------------------|---------|--------|----------|--------|-----------|--------|-----------|--------|---------|
|                           |         | dips   | unbaited | baited | Unbaited  | baited | unbaited  | baited | landing |
| <b>Number Equivalents</b> |         |        |          |        |           |        |           |        |         |
| <i>. albopictus</i>       | 100     |        | 3        | 8      | 1         | 3      | 1         | 3      | 10      |
| <i>. caspius</i>          |         |        |          | 10     | 5         | 100    | 1         | 40     | 50      |
| <i>. japonicus</i>        | 15      |        | 1        | 2      |           |        | 1         | 2      | 3       |
| <i>. vexans</i>           |         |        | 5        | 10     | 5         | 100    | 1         | 40     | 50      |
| <i>. pipiens</i>          |         | 5      | 5        | 10     |           | 50     |           | 20     | 6       |
| <i>thelleri</i>           |         | 5      |          | 10     |           |        |           | 20     |         |
| <b>Conversion Factors</b> |         |        |          |        |           |        |           |        |         |
| <i>. albopictus</i>       | 1       |        | 33.3     | 12.75  | 100       | 33.3   | 100       | 33.3   | 10      |
| <i>. caspius</i>          |         |        |          | 4      | 8         | 0.4    | 40        | 1      | 0.8     |
| <i>. japonicus</i>        |         |        | 2        | 1      |           |        | 2         | 1      | 0.67    |
| <i>. vexans</i>           |         |        | 8        | 4      | 8         | 0.4    | 40        | 1      | 0.8     |
| <i>. pipiens</i>          |         | 4      | 4        | 2      |           | 0.4    |           | 1      | 3.3     |
| <i>thelleri</i>           |         | 4      |          | 2      |           |        |           | 1      |         |

## Aggregation to de-duplicate:

Easy for for PA

1, 5 or 10km. 1 or 10 easiest as can use lat long decimal number  
can be by date as well: annual abundance aggregated as Maximum

Less easy for abundance: standardised max ( or ?mean) per sample unit per  
date

Can be done by date/month

## Absences for PA:

Not pseudo absence - unless certain that sampling is complete  
should be via suitability thresholds – to allow absence near areas of presence